An object thrown at time $t = 0$ from initial height s_0 with initial velocity v_0 satisfies the equation.

$$a = -g, \quad v = -gt + v_0, \quad s = -\frac{gt^2}{2} + v_0 t + s_0 .$$

where, $g = 9.8 \text{ m/s}^2$ or 32 ft/s^2.

A few examples of **quantities** which are the **rates of change with respect to some other quantity** in our daily life are given below:

1. Slope is the **rate of change in vertical length with respect to horizontal length.**
2. Velocity is the **rate of displacement with respect to time**.
3. Acceleration is the **rate of change in velocity with respect to time**.
4. The steepness of a hillside is the **rate of change in its elevation with respect to linear distance**.

Consider the following two situations:

- A person is continuously driving a car from Chennai to Dharmapuri. The distance (measured in kilometre) travelled is expressed as a function of time (measured in hours) by $D(t)$. What is the meaning one can attribute to $D'(3) = 70$?

  It means that, **"the rate of distance when $t = 3$ is 70 kmph"**.

- A water source is draining with respect to the time t. The amount of water so drained after t days is expressed as $V(t)$. What is the meaning of the slope of the tangent to the curve $y = V(t)$ at $t = 7$ is -3?

  It means that, **"the water is draining at the rate of 3 units per day on day 7"**.

Likewise the rate of change concept can be used in our daily life problems. Let us now illustrate this with more examples.

Example 7.2

The temperature T in celsius in a long rod of length 10 m, insulated at both ends, is a function of length x given by $T = x(10 - x)$. Prove that the rate of change of temperature at the midpoint of the rod is zero.

Solution

We are given that, $T = 10x - x^2$. Hence, the rate of change at any distance from one end is given by $\frac{dT}{dx} = 10 - 2x$. The mid point of the rod is at $x = 5$. Substituting $x = 5$, we get $\frac{dT}{dx} = 0$. ■

Example 7.3

A person learnt 100 words for an English test. The number of words the person remembers in t days after learning is given by $W(t) = 100 \times (1 - 0.1t)^2$, $0 \le t \le 10$. What is the rate at which the person forgets the words 2 days after learning?

Solution

We have,

$$\frac{d}{dt}W(t) = -20 \times (1 - 0.1t).$$

Therefore at $t = 2$, $\frac{d}{dt}W(t) = -16$.

That is, the person forgets at the rate of 16 words after 2 days of studying. ■

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